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MILITARY SPECIFICATION

ELECTRICAL SYSTEMS, AIRCRAFT, DESIGN AND INSTALLATION OF, GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for the design and installation of alternating-current and direct-current electrical systems in military aircraft (see 6.3.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-7016	Electric Load and Power Source Capacity, Analysis of, Method for Aircraft and Missiles
MIL-I-7032	Inverter, Aircraft, General Specification for
MIL-E-7080	Electric Equipment, Piloted Aircraft Installation and Selection of, General Specification for
MIL-E-24021	Electric Power Monitor, External, Aircraft
MIL-F-25381	Flight Testing, Electric System, Piloted Aircraft and Guided Missile, General Requirements for
MIL-M-25500	Mockup Testing, Electric System, Piloted Aircraft and Guided Missile, General Requirements for
MIL-E-25704	Generator System, Alternating Current, Aircraft, General Specification for
MIL-I-27273	Inverter, Power, Static, General Specification for

FSC MISC

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STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods
MS17793	Schematic Wiring Diagram (External AC Power Connector, Aircraft)
MS24435	Connector Receptacle, Power Outlet, Electronic Test Equipment
MS24439	Wiring Diagram - Battery Circuit, Schematic
MS24442	Wiring Diagram - External Power Receptacle, 28 Volt D.C., Common Bus System

PUBLICATION

Air Force Manual

AFSCM/AFLCM 310-1 Management of Contractor Data and Reports

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements

3.1.1 Submission of design data. Design data in accordance with AFSCM/AFLCM 310-1 shall be submitted to the procuring activity.

3.1.2 Deviations. Deviations to the requirements of this specification shall not be made without approval of the procuring activity.

3.1.3 Reliability. The reliability required of the electrical installation and system design shall be that which is necessary to achieve the required reliability of the overall weapon system. The installation shall provide for standardization and simplicity of operation as long as reliability is not affected adversely. Reliability requirements shall be as specified in the detail specification.

3.1.4 Maintenance. Ease of maintenance shall be consistent with equipment life and maintenance requirements. A positive means such as polarization shall be provided on power-system equipment terminals to prevent incorrect connection of wiring.

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Insofar as practicable, parts of the equipment shall be made readily accessible for servicing without teardown of the equipment and removal of any major part, component, or accessory. Maintainability requirements shall be as specified in the detail specification.

3.1.5 Assembly. Insofar as practicable, the following general requirements shall be adhered to in order to reduce the possibility of human errors during assembly and maintenance of system components:

3.1.5.1 Hardware. Mounting hardware shall be permanently connected to the part being mounted.

3.1.5.2 Stacked assemblies. Stacked assemblies shall use flanged arrangements that prevent assembly if a part is missing. Alternatively, indicator markings which are exposed, if parts are omitted, may be used.

3.1.5.3 Right- and left-hand parts shall be identical and interchangeable or shall have some design feature to prevent their being interchanged during assembly.

3.1.5.4 Bolts. When bolts of different sizes are used near the same location, widely differing sizes shall be specified. Two bolts that are identical except for length shall not be used in the same location. Design shall protect against damage to parts due to accidental use of screws or bolts that are too long.

3.1.5.5 Mounting. Where reversed or rotated mounting of a part cannot be tolerated, nonsymmetrical mounting arrangements (including keyways or pins) shall be used. Parts that are capable of proper operation when mounted in any orientation are more desirable.

3.1.5.6 Fluid fittings. Where fluid fittings are required, maximum use shall be made of the different sizes of standard fluid fittings available.

3.1.6 Installation. Electric equipment shall be installed in accordance with MIL-E-7080 and as specified herein. The requirements specified herein shall govern in case of conflict.

3.2 Detail requirements. The following detail requirements shall apply to specific items of equipment or systems.

3.2.1 Wiring. All wiring, conduit, junctions, and shielding shall be installed in accordance with MIL-W-5088 except where that specification conflicts with the following:

3.2.1.1 Generator wiring. The generator wiring shall be installed as follows:

a. Sufficient slack shall be provided to prevent generator movement from placing

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damaging stress on the generator terminal block and to allow for terminal block location of 6 to 12 inches from the mounting flange

b. The generator field and power leads shall be routed in different bundles. The field leads shall be routed and protected to minimize unintentional connection to any power circuit

c. Where the aircraft structure is used for system neutral or negative return, the neutral or negative cable of the generator shall be grounded to the nearest main structure capable of satisfactorily carrying the current, after passing through the negative shunt or transformer if installed

d. Where a neutral or negative cable return is used due to high resistance structure or to eliminate ground loops and signal interference, the return shall be grounded to the structure at one point only, as near the generator as practicable. The neutral return and ground point shall be capable of carrying the short circuit current of the generating system.

3.2.2 Electric power systems. Figure 1 establishes the terminology for the electric system as applicable to this specification. The terminology contained on figure 1 is not intended to establish design philosophy.

3.2.2.1 System characteristics. Electrical systems characteristics shall be in accordance with MIL-STD-704.

3.2.2.2 System generating capacity

3.2.2.2.1 Single engine. Single-engine aircraft shall have a primary generator system which has a continous kilovolt-ampere capacity at least 33 percent greater than the maximum continuous electrical load (see 6.3.2) of the initial production aircraft.

3.2.2.2.2 Multiengine. Multiengine aircraft shall have a multigenerator primary electrical system which has a continuous kilovolt-ampere capacity of at least twice the maximum continuous electrical load (see 6.3.2) of the initial production aircraft.

3.2.2.2.3 Electrical load analyses. Electrical load analyses in accordance with MIL-E-7016 shall be furnished as specified by the data requirements of the contract.

3.2.2.3 Point of voltage regulation. The voltage shall be regulated at a point as near as practicable to the main bus. The voltage at this point shall be adjusted to a nominal value of 28V line-to-ground for d-c systems and 115V line-to-neutral for a-c systems.

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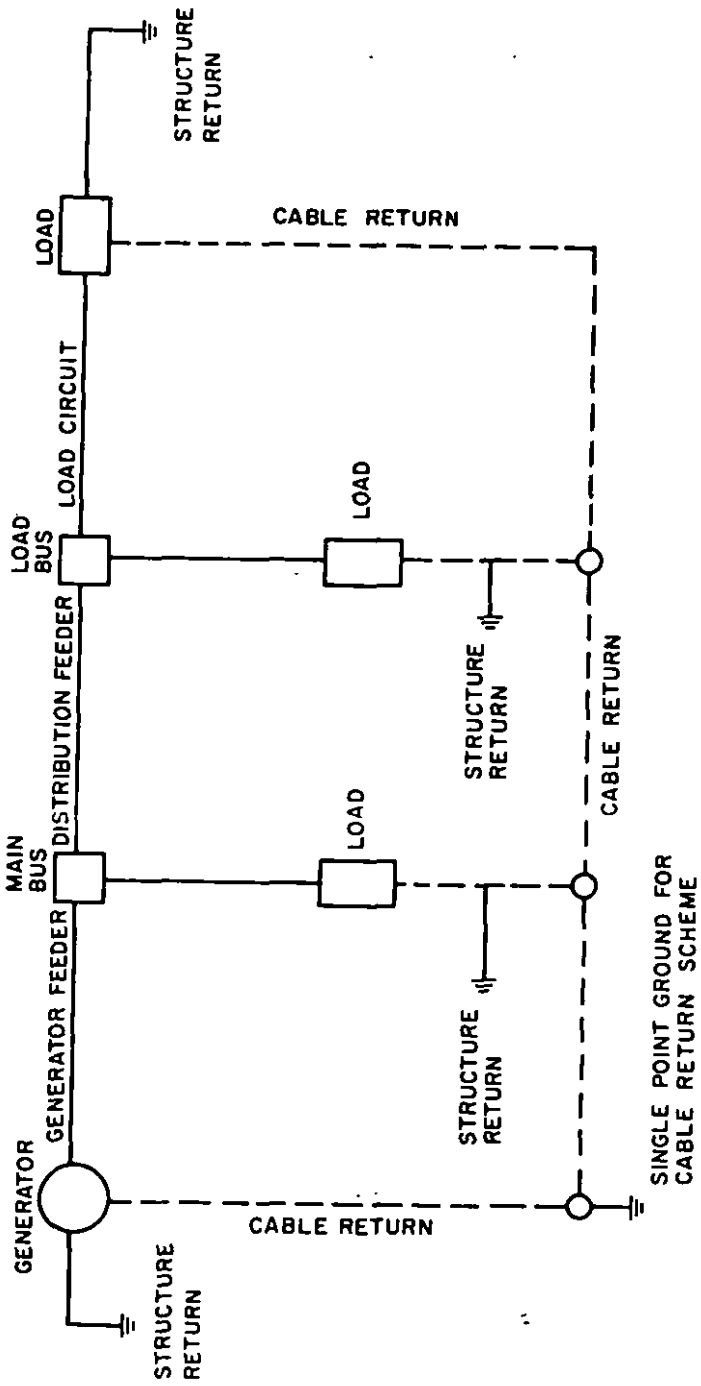


FIGURE 1. Electrical System Terminology Chart

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3.2.2.4 Direct-current power systems

3.2.2.4.1 Generator system

3.2.2.4.1.1 Operator's controls. As a minimum requirement, a control switch and a generator trip indicator shall be installed for each generator. These controls shall be grouped together on the engineer's, pilot's, or copilot's electrical panel, as applicable. Control grouping shall be consistent with system operational and reliability requirements. Voltmeters, loadmeters, and associated switches shall be provided as required for safe and effective management of the generator system.

3.2.2.4.2 Battery system. The battery circuit shall conform to MS24439, or shall be a circuit having equal or better performance.

3.2.2.4.2.1 Battery relay. A relay shall be installed in each battery circuit to isolate the battery from the normal system, controlled by a switch installed on the pilot's or engineer's electrical switch panel. The relay shall not be connected through the ignition switch, and shall be located as close as practicable to the battery.

3.2.2.4.3 Transformer-rectifier power. Transformer-rectifiers shall be in accordance with applicable detail specifications. Transformer-rectifier direct-current systems shall employ fused multichannel circuits to provide short-circuit protection and greater system reliability. Another type of short-circuit protection may be used provided it has equal or better performance. Where more than one transformer-rectifier is used, the units shall be so distributed among the various alternating-current buses as to obtain the most reliable operation. When two or more transformer-rectifier units supply a common direct-current bus, system requirements for load sharing shall be consistent with the allowable voltage tolerances and regulation characteristics of the individual units.

3.2.2.5 Alternating-current power systems

3.2.2.5.1 A-C generator systems. A-C generator systems shall be in accordance with MIL-E-25704.

3.2.2.5.1.1 Operator's controls (ac). As a minimum requirement, a generator control switch and a generator breaker indicator shall be installed for each generator. In addition, for parallel operation, one bus-tie breaker control switch and one bus-tie breaker indicator shall be provided for each generator. These controls shall be grouped together on the engineer's, pilot's, or copilot's electrical panel as applicable. Control grouping shall be consistent with system operational and reliability requirements. Voltmeters, loadmeters, and associated switches shall be provided as required for safe and effective management of generator systems.

3.2.2.5.2 Inverter power system. Static inverters shall be in accordance with MIL-I-27273 and rotary inverters shall be in accordance with MIL-I-7032.

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3.2.2.5.2.1 Inverter changeover. When equipment that is essential for maintaining controlled flight derives power from an inverter, a spare inverter shall be provided. Chaneover from the main inverter to the spare shall be automatic in the event of main inverter failure. Spare inverters may also be used as operative spares to supply power to nonessential buses but the nonessential bus shall be dropped in the event of main inverter failure.

3.2.2.6 Direct- and alternating-current distribution systems

3.2.2.6.1 Distribution feeders, alternating- and direct-current. The distribution feeders shall consist of the proper number and size of wires to satisfy system characteristics and reliability requirements. In multiple-wire feeders, if one wire opens, the remaining intact wires shall not be loaded in excess of 125 percent of rated current, based on the current-carrying capacities specified in MIL-W-5088 or the appropriate wire specification. The ratio of the lowest resistance to the highest resistance of any parallel paths (including wire and terminals) shall be as near unity as practical but not less than 0.9. Where fuse-protected feeders (a fuse or equivalent protector at each end of each wire) are installed, provisions shall be made to insure proper coordination of fuses. Fuses shall not be paralleled by jumpering the terminals at each end of the fuse block, but each fuse shall have a sufficient length of wire in series to provide at least 0.01 ohm resistance.

3.2.2.6.2 Channel separation. The distribution feeder wires in a direct-current system, or phase groups in an alternating-current system, shall be so located as to reduce vulnerability of the system to a minimum. For example, in a three-wire direct-current system, the first wire could be routed along the left side of the fuselage, the second wire along the top-center portion of the fuselage, and the third wire along the right side of the aircraft. The phase groups in an alternating-current system should be similarly routed.

3.2.2.6.2.1 Minimum separation. Where applicable, the minimum separation between direct-current wires or alternating-current phase groups shall be 3 inches, except where entering junction boxes, to reduce vulnerability of the system due to gunfire.

3.2.2.6.3 Rigid bus bar. When a rigid bus bar is used, its length shall be kept to a minimum and it shall be installed in a junction box or otherwise suitably protected.

3.2.2.6.4 Bus system. The bus system shall be configured to safely carry all connected loads. Buses may be segmented to various areas of the airplane as required to achieve an optimum distribution system. The bus system shall be configured in such a manner that under normal conditions, all loads will receive power from the primary power source. Essential loads shall have first priority to the available power. Provision shall be made to automatically disconnect nonessential loads as required in the event of an electrical system failure which would reduce the amount of available power below the total requirements of the aircraft.

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3.2.2.6.5 Main buses. The main buses shall be as short as possible and shall be located in individual junction boxes to reduce vulnerability. In locating these boxes, advantage shall be taken of the protection offered by the structural members of the aircraft whenever possible. The aircraft loads connected to each main bus shall be as nearly equal as possible. Cables for interconnecting individual buses shall consist of multiple wires with suitable electrical protection to ensure that a fault on any wire will not affect the integrity of the bus. The individual bus tie wires shall be physically separated by the greatest distance practicable throughout their length.

3.2.2.7 Electromagnetic interference. The individual electrical equipments shall meet the electromagnetic interference requirements of MIL-STD-461.

3.2.3 Emergency power source. An independent source of emergency power shall be provided to supply power to all essential loads in the event of failure of the normal power source.

3.2.4 Generator oil system. A cooling and lubricating oil system independent of the oil systems for the engine and other components shall be provided for oil-cooled generators and hydraulic constant-speed drives.

3.2.5 External power

3.2.5.1 Direct-current receptacles. For aircraft having a common bus, the external power receptacle or receptacles shall be installed in accordance with MS24442.

3.2.5.2 Alternating-current receptacles. The external power receptacle, or receptacles, shall be installed in accordance with MS17793. Transfer of aircraft loads from external power to on-board power and from on-board power to external power shall be accomplished in a manner consistent with the transients and power interruptions which can be tolerated by the aircraft loads.

3.2.5.3 Voltage drop. The voltage drop from the external power receptacle to the load shall be such that equipment operation is above the design low-voltage limits when minimum rated external voltage is applied at the receptacle. For electric starters, nominal external voltage shall be used for design.

3.2.5.4 Automatic release. For interceptor-type aircraft, the external power receptacles shall be installed to permit automatic release of the external power cables upon movement of the aircraft where manual disconnect would create a safety hazard or impair alert requirements.

3.2.5.5 External power protection. An external power monitor in accordance with MIL-E-24021 shall be provided to prevent unsuitable external power from being applied to the aircraft.

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3.2.6 Relays and contractors. All relays and the interlock switches of circuit breakers and contactors shall be hermetically sealed. Unsealed relays shall not be used.

3.2.7 Switches and circuit breakers

3.2.7.1 Automatic reset breakers. Automatic reset breakers shall be used only when specifically approved by the procuring activity. When used, they shall be in conjunction with switches, control relays, or other circuit-opening means.

3.2.8 Auxiliary power units. When auxiliary, engine-driven power plants are required, the electric control circuit shall be subject to approval of the procuring activity. The voltage regulator of d-c power plants shall be so adjusted that a nominal voltage of 27V is obtained. A placard with the following instructions shall be permanently installed in the vicinity of the voltage regulator:

AUXILIARY POWER UNIT VOLTAGE REGULATOR: TO BE ADJUSTED BY QUALIFIED PERSONNEL ONLY. DO NOT ADJUST ABOVE 27V.

3.2.9 Test equipment receptacles. Receptacles to the supply power for test equipment shall be installed in accordance with MS24435.

3.3 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.4 Workmanship. All details of workmanship shall be in accordance with high-grade aircraft electrical installation practices. Adequate quality control shall be exercised in the installation of the systems and in production workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Tests. Generator system tests shall be conducted in accordance with MIL-E-25704; mockup tests shall be in accordance with MIL-E-25500; flight tests shall be in accordance with MIL-F-25381; environmental testing on electrical equipment shall be in accordance with MIL-STD-810; and electromagnetic interference tests shall be in accordance with MIL-STD-462.

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5. PREPARATION FOR DELIVERY

5.1 Section 5 is not applicable to this specification.

6. NOTES

6.1 Intended use. The requirements covered by this specification are intended for use in designing and installing electrical systems in aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Submission of design data (see 3.1.1 and 3.1.2)

c. Submission of electrical load analyses (see 3.2.2.2.3).

6.3 Definitions

6.3.1 Electrical systems. The term electric systems means electric power generation and distribution circuits, including control and protective devices.

6.3.2 Maximum continuous electrical load. The maximum continuous electrical load is the largest load which exists for an operational period of 15 minutes or longer based on the load analysis.

6.4 Identification of international standardization agreements. Certain provisions (see 3.2.5.1, 3.2.5.2, and 3.2.7.1) of this specification are the subject of international standardization agreement(ASCCAS 12/6, paragraph b, and STANAG 3302 apply to 3.2.5.1; ASCCAS 12/7, paragraph b, and STANAG 3303 apply to 3.2.5.2 and ASCCAS 12/3 applies to 3.2.7.1). When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.5 Marginal indicia. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AV
Air Force - 11

Review activities:
Army - EL
Air Force - 70

Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 65 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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